

Unit 2: Advanced Features in Spreadsheet

Topics Covered: Array Formula for Complex Calculation; Advanced Financial Function (SLN, DB and DDB, PMT, PPMT, IPMT, IRR, NPV and XIRR, XNPV, ISPMT)

Assignment 5

The objective of this Spreadsheet assignment is to enhance students' proficiency in utilizing array formulas for complex calculations and to deepen their understanding of advanced financial functions. Students will learn to efficiently apply array formulas to solve multifaceted business data problems, optimizing their analytical capabilities. Additionally, the assignment will cover a comprehensive range of advanced financial functions. Mastery of these functions will enable students to perform sophisticated financial analyses, such as calculating depreciation, loan payments, interest, and investment returns, thereby equipping them with essential skills for real-world financial decision-making and data-driven insights.

Open a blank workbook and save it as Spreadsheet5_rollno.xlsx in your folder.

Array Formula for Complex Calculation

1. In array_rollno sheet, add the following data.

Item	Units Sold	Unit Price
Apple	10	40
Kiwi	30	85
Lemon	20	25
Mango	30	50
Peach	40	80

- a. Calculate grand total of all sales using array formula.
- b. Suppose you have to pay 10% tax for each sale. Calculate the tax amount for each item using array formula.

Advanced Financial Functions

2. Cozy Corner Interiors possesses an asset worth 850000 INR. The Salvage Value is 135000 INR, and the Expected Life of the asset is assumed to be 9 years.
 - a. Calculate the annual depreciation to be charged for the asset using Straight Line method. Also calculate how much provision is to be created monthly for the Depreciation (Hint: SLN).
 - b. Consider the same asset and prepare the annual depreciation tables using Fixed Declining Balance method and Double Declining Balance method over the life of the asset (Hint: DB, DDB).
3. Ms. Anita Sharma, a cake shop owner, plans to purchase a new home. The property is priced at 60 lakhs INR. She decided to pay 20% of the total amount as down payment. She approached a finance company for Home Loan with 15% interest rate for a tenure of 10 years.

ST. XAVIER'S COLLEGE (AUTONOMOUS), KOLKATA
Department of Information Technology
B. Com Semester III
IT Skills for Business Practical 2024 under NEP 2020

- a. Calculate the amount of down payment and monthly repayment (EMI) to repay the loan (Hint: PMT).
 - b. Calculate the principal portion of the first month's payment (Hint: PPMT).
 - c. Calculate the interest portion of the first month's payment (Hint: IPMT).
4. Lifeline Pharma started online business in the year 2018. The following cash flow scenario has been observed.

Year	0	1	2	3	4	5
Cashflow	-100000	20000	25000	30000	35000	40000

Calculate Internal Rate of Return and Net Present Value (Hint: IRR, NPV).

5. Consider the following irregular cash flows for a project.

Year	01-10-2021	02-02-2022	04-08-2023	07-10-2024	11-11-2025
Cashflow	-100000	30000	25000	30000	30000

- a. Calculate Extended Internal Rate of Return of the project (Hint: XIRR).
 - b. Calculate Extended Net Present Value of the project (Hint: XNPV).
6. Mr. Rajeev Bhatia wants to take a loan of 800000 INR for 4 years. The rate of interest is 10% p.a. Prepare Even Principal or Straight-Line Loan Amortization Table (Yearly) using ISPMT [Period, Principal, Interest, Payment, Balance].

Reference Links:

[https://support.microsoft.com/en-us/office/guidelines-and-examples-of-array-formulas-7d94a64e-3ff3-4686-9372-ecfd5caa57c7#ID0EBB\]=Windows](https://support.microsoft.com/en-us/office/guidelines-and-examples-of-array-formulas-7d94a64e-3ff3-4686-9372-ecfd5caa57c7#ID0EBB]=Windows)
<https://support.microsoft.com/en-us/office/financial-functions-reference-5658d81e-6035-4f24-89c1-fbf124c2b1d8>
